# COMMUNICATION IN ENGINEERING

*Bad Communication vs. Good Communication*

*3rd-Cycle Engineering · English Module*

## 1 · Introduction

Communication is the process of exchanging information, ideas, and feelings between individuals or groups. In engineering contexts, effective communication is critical for project success, team collaboration, and professional growth. This document contrasts **bad communication** and **good communication**, covering definitions, types, barriers, advantages, disadvantages, and real-world examples.

> *"The single biggest problem in communication is the illusion that it has taken place." — G.B. Shaw*

## 2 · Types of Communication

| Type | Description | Examples |
|---|---|---|
| Verbal | Spoken words, tone, and voice | Meetings, calls, presentations |
| Written | Text-based exchange | Emails, reports, manuals |
| Non-verbal | Body language, gestures, facial expressions | Nodding, posture, eye contact |
| Visual | Graphs, diagrams, charts | Technical drawings, infographics |
| Digital | Electronic & online channels | Slack, video calls, social media |

## 3 · Bad Communication vs. Good Communication

| ✗ BAD COMMUNICATION | ✓ GOOD COMMUNICATION |
|---|---|
| • Vague or ambiguous language | • Clear, precise, and concise |
| • Ignoring the audience's level | • Adapted to the audience |
| • Poor listening / interrupting | • Active listening & empathy |
| • Emotional / aggressive tone | • Respectful and professional tone |
| • Information overload | • Structured and well-organised |
| • Lack of feedback or follow-up | • Timely feedback and follow-up |
| • Using jargon inappropriately | • Appropriate technical vocabulary |
| • Wrong channel chosen | • Right channel for the message |
| • Cultural insensitivity | • Culturally sensitive |
| • No clear structure or purpose | • Goal-oriented with clear intent |

## 4 · Advantages & Disadvantages

| GOOD COMMUNICATION — Advantages | BAD COMMUNICATION — Consequences |
|---|---|
| ✔ Increases team productivity and efficiency | ✘ Leads to misunderstandings and errors |
| ✔ Builds trust and stronger relationships | ✘ Creates conflicts and low team morale |
| ✔ Faster problem-solving and decision-making | ✘ Delays in project delivery |
| ✔ Reduces costly rework and mistakes | ✘ Increases costs due to repeated errors |
| ✔ Promotes innovation through idea-sharing | ✘ Stifles creativity and initiative |
| ✔ Enhances professional reputation | ✘ Damages credibility and career growth |
| ✔ Clear project specs reduce ambiguity | ✘ Vague specs cause scope creep |

## 5 · Communication Barriers

| Barrier | Cause | Solution |
|---|---|---|
| Physical | Noise, distance, poor infrastructure | Use video calls, written backups |
| Language | Technical jargon, foreign language | Simplify, use visuals, glossaries |
| Psychological | Stress, bias, lack of confidence | Foster open, safe environment |
| Cultural | Different norms and values | Cross-cultural training |
| Organisational | Hierarchy blocking information | Flat communication structures |
| Technological | Tool failures, digital divide | Train staff, have backup plans |

## 6 · The 7 C's of Effective Communication

**Clear**
Express ideas without ambiguity

**Concise**
Use only necessary words; avoid padding

**Correct**
Ensure factual and grammatical accuracy

**Complete**
Provide all required information

**Coherent**
Logical flow and consistency of ideas

**Courteous**
Respectful and considerate tone

**Concrete**
Specific facts, figures, and examples

## 7 · Real-World Engineering Examples

| Scenario | Bad Communication | Good Communication |
|---|---|---|
| Design Review | Engineer says: 'The system might have issues.' → vague, no action taken | Engineer says: 'Valve X exceeds pressure limit by 15%. Replace before testing.' |
| Team Meeting | Manager interrupts constantly, team afraid to speak up → hidden risks | Manager listens actively, encourages questions → all risks identified early |
| Client Report | Report full of jargon, no summary → client misunderstands deliverable | Report has executive summary, clear charts → client approves promptly |
| Crisis / Incident | Alarm ignored due to unclear protocol → accident escalates | Clear emergency procedure communicated → swift controlled response |

## 8 · Conclusion

Effective communication is not a soft skill — it is a **core engineering competency**. Bad communication leads to errors, conflicts, delays, and safety risks. Good communication, grounded in clarity, empathy, and structure, drives project success, team cohesion, and professional excellence. As future engineers, mastering communication in both technical and interpersonal contexts is essential for a successful career.